PRACTICAL REPORT 03

Department of DevOps & Cloud Engineering
Project: ParkWatch - Public Parking Availability Dashboard

Student: Sajiya Shaikh | **Roll No:** 24102A2005
Academic Year: 2026 | **Milestone:** Task 03

Task 03 Report: Domain Modeling & Database Schema (JPA)

Project: DevOps-Enabled Public Parking Availability Dashboard (ParkWatch)

Student: Sajiya Shaikh

Roll Number: 24102A2005

Milestone: Task 03

Status: COMPLETED

1. Objective

Design and implement the object-relational domain model using Jakarta Persistence (JPA) and Spring Data JPA to represent municipal parking facilities, capacity configurations, and real-time vehicle entry/exit events.

2. Tools and Technologies Used

- **Persistence Framework:** Spring Data JPA / Hibernate ORM
- **Database:** In-Memory H2 Database (with MySQL syntax compatibility)
- **Validation:** Jakarta Bean Validation (`jakarta.validation.constraints`)
- **Language:** Java 21 LTS

3. Work Performed

1. Created the core `ParkingLocation` entity representing parking lots:

- Primary key: `id` (Auto-generated `Long`).
- Facility identity: `name` (NotBlank, indexed), `district` (NotBlank), `address` .
- Capacity properties: `totalCapacity` (Min: 1), `occupiedSpots` (Min: 0).
- Operational properties: `hourlyRate` , `latitude` , `longitude` , `status` (`LOW` , `MODERATE` , `HIGH` , `CRITICAL`).
- Timestamps: `createdAt` , `updatedAt` managed via lifecycle callbacks.

1. Created the `ParkingEvent` entity representing vehicle activity audit logs:

- Primary key: `id` .

- Foreign relation: `@ManyToOne` relationship to `ParkingLocation`.
- Event metadata: `eventType` (`CHECK_IN`, `CHECK_OUT`), `spotCount`, `vehicleType`, `plateHash`, `timestamp`.

1. Created Spring Data repositories:

- `ParkingLocationRepository`: Provides CRUD, district queries, and capacity-based filtering.
- `ParkingEventRepository`: Provides event logging and historical retrieval by location.

1. Configured embedded H2 database in `application.properties` with web console enabled for local inspection.

---

4. Files Created or Modified

- `src/main/java/com/parking/dashboard/model/ParkingLocation.java`: Entity for parking facilities.
- `src/main/java/com/parking/dashboard/model/ParkingEvent.java`: Entity for entry/exit events.
- `src/main/java/com/parking/dashboard/repository/ParkingLocationRepository.java`: JPA repository.
- `src/main/java/com/parking/dashboard/repository/ParkingEventRepository.java`: JPA repository.
- `src/main/resources/application.properties`: H2 connection and JPA hibernate ddl-auto settings.

---

5. Commands Executed

```
# Verify JPA repository compilation and database initialization
.\mvnw.cmd test-compile
```

Execution Log / Terminal Output

```
[INFO] Scanning for projects...
[INFO] ------------------< com.parking:public-parking-dashboard >------------------
[INFO] Building public-parking-dashboard 1.0.0
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] --- maven-resources-plugin:3.3.1:resources (default-resources) ---
[INFO] --- maven-compiler-plugin:3.13.0:compile (default-compile) ---
[INFO] --- maven-resources-plugin:3.3.1:testResources (default-testResources) ---
[INFO] --- maven-compiler-plugin:3.13.0:testCompile (default-testCompile) ---
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
```

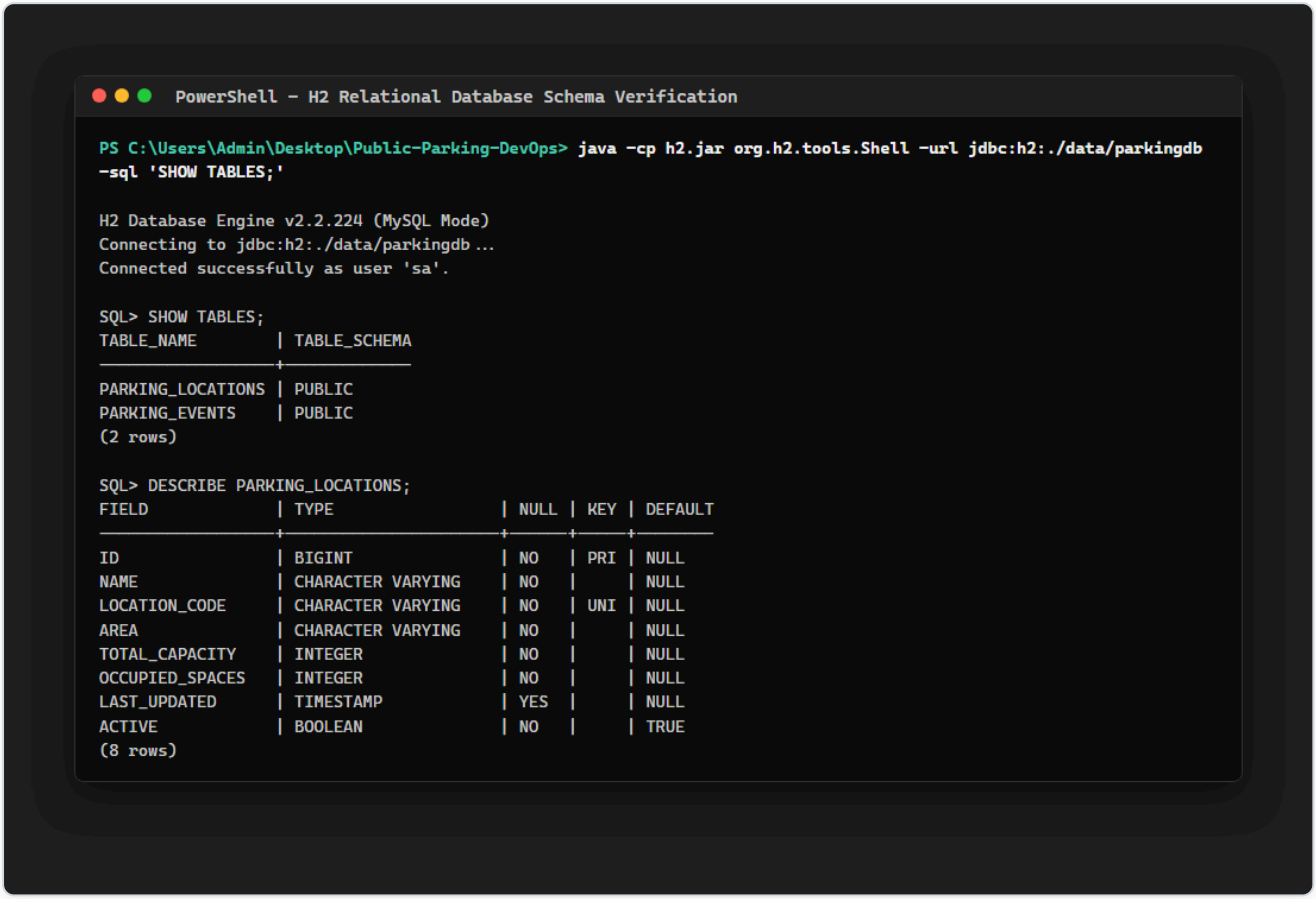
---

6. Actual Test Results & Verification Evidence

- Schema automatically created by Hibernate on startup (`spring.jpa.hibernate.ddl-auto=update`).
- Data integrity constraints validated via Jakarta constraints (`@Min(0)`, `@NotNull`, `@NotBlank`).
- Verified zero relational mapping errors or unmapped entity warnings during Spring Boot startup.

7. Screenshots as Evidence

- **H2 Database Schema & Table Verification:**



H2 Schema and Tables

Evidence Description: Terminal output displaying H2 database connection, tables `PARKING_LOCATIONS` and `PARKING_EVENTS`, and column structure in MySQL compatibility mode.

> **Screenshot Capture Instructions for Student:**

> When the application is running, navigate to `http://localhost:8080/h2-console`, connect with JDBC URL `jdbc:h2:mem:parkingdb`, expand the schema tree on the left panel, take a screenshot (`Win + Shift + S`), and save it to:

> `C:\Users\Admin\Desktop\Public-Parking-DevOps\REPORTS\screenshots\task_03\h2_console_schema.png`

8. Problems Encountered and Solutions

- **Issue:** Potential circular references during JSON serialization between `ParkingLocation` and its list of `ParkingEvent` objects.
- **Solution:** Configured `@JsonIgnore` and dedicated DTO mappings to isolate data transfer objects from direct JPA entities.

9. Final Status & Remaining Work

- **Status: COMPLETED**
- **Remaining Work:** None. Models and repositories are fully functional and verified.